Mathematics for Scientists

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Mathematics for Scientists (A11522)

Short Title:	Mathematics for Scientists
Faculty	Science and Computing
Department	Science
Cluster	Mathematics and Physics
Author	AVEREKER
Credits	5

Level: Introductory

Description of Module / Aims

This module will explore the basics of differentiation, concentrating on rates of change and optimisation. It will also introduce the student to the idea of areas of irregular shapes and how integration can be used to calculate these. There is an emphasis throughout this module on the role of calculus in the sciences, including real world applications.

Programmes

MTHS-0002	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / M
MTHS-0002	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 2 / M
MTHS-0002	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / M
MTHS-0002	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / M
MTHS-0002	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 2 / M
MTHS-0002	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 2 / M
MTHS-0002	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / M
MTHS-0002	BSc in Food Science with Business (WD_SFDSC_D)	1 / 2 / M
MTHS-0002	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 2 / M
MTHS-0002	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 2 / M

Indicative Content

- Introduction to differentiation: finding the limits and slopes of polynomial, rational, exponential, logarithmic and trigonometric functions through the use of first principles
- Differentiation: using appropriate methods of differentiation to find the first derivative and second derivative of a range of functions
- Applications of differentiation: rates of change; critical points and optimisation
- Introduction to integration: anti-derivatives; area under a curve; area between two curves; trapezoidal rule; methods of integration
- Applications of integration: calculate the total and average values of various functions over a specific range of values and introduce differential equations

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use appropriate differentiation methods to differentiate linear, polynomial, rational, exponential and trigonometric functions.
2. Use differentiation to solve elementary optimisation problems.
3. Use appropriate integration methods to integrate linear, polynomial, rational, exponential and trigonometric functions.
4. Calculate irregular areas using integration techniques.
5. Calculate irregular areas using numerical methods.
6. Apply differentiation and integration to solve mathematical problems.
7. Interpret results obtained from problems of an applied nature in the science environment.

Learning and Teaching Methods

- Lectures.
- Online exercises.
- Use of technology to enhance the learning experience.
- Self-directed learning is emphasised. Students are supported with notes, online material and feedback.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	5,6,7
Continuous Assessment	30%	
In-Class Assessment	10%	1,2,3,4
Tutorial/Problem Sheets	20%	1,2,3,4

Assessment Criteria

<40%: Inability to identify and describe key concepts of calculus.

40%-49%: Ability to identify and describe key concepts of calculus.

50%-59%: Ability to discuss key concepts of calculus clearly and interpret their relative importance.

60%-69%: Ability to apply solutions to problems in a range of relevant contexts by using differentiation and integration. Ability to employ a comprehensive range of specialised calculus skills.

70%-100%: All the above to an excellent level, with the ability to analyse and design solutions to a high standard for a range of complex or unseen problems using calculus.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Croft, A. and R. Davison. _Foundation Maths_. 5th ed. England: Pearson, 2010.

Supplementary Material

- Bird , J.O. and A.J.C. May. _Technician Mathematics 1_. 3rd ed. Malaysia: Longman, 1994.
- Bird, J.O. and A.J.C. May. _Technician Mathematics 2_. 3rd ed. Malaysia: Longman, 1994.
- Bird, J.O. and A.J.C. May. _Technician Mathematics 3_. 3rd ed. Malaysia: Longman, 1994.
- Thomas, G.B. and Finney. _Calculus_. 8th Edition. New Jersey: Addison Wesley, 1992.

Requested Resources

- Room Type: Computer Lab